REDUX TOOLKIT (RTK) vs ZUSTAND

BASICS COMPARISON

★ 1. What are they?

- RTK → Official, opinionated, batteries-included state management for large-scale apps.
- Zustand → Small, simple, unopinionated store for React. Uses hook-based API.

Both Manage State ! ♡

★ 2. Core Philosophy

Redux Toolkit (RTK)	Zustand
Follow structure and patterns Best for predictability and maintainability.	Minimal API Do what you need, no extra rules.

★ 3. Boilerplate Code

RTK → More setup

- Store
- Slice
- Actions
- Provider

Zustand → Very less setup

- Create store
- Use store in component

★ 4. Setup Example

RTK Setup Flow

createSlice()
↓
configureStore()
↓
<Provider store={store}>
↓
useSelector / useDispatch()

Zustand Setup Flow

create()
↓
useStore()
↓
Done ✓

★ 5. State Access in Components

RTK

```
const value = useSelector(
  (state) → state.slice.value
)
```

```
const dispatch = useDispatch()
```

Zustand

```
const value = useStore(
  (state) → state.value
)
```

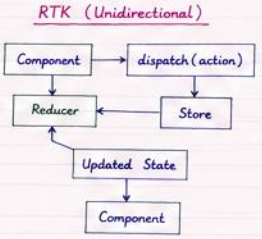
Direct hook usage ! No dispatch needed for simple updates.

☺

★ 6. State Update

RTK	Zustand
dispatch(increment()) dispatch(setValue(10)) Updates are done by dispatching actions.	const increment = useStore((state) → state.increment) increment() Call function directly from the store.

★ 7. Data Flow



★ 8. Best For

RTK	Zustand
Large applications	Small to medium applications
Complex state logic	Simple state logic
Teams who like structure	Teams who like simplicity and flexibility
Need devtools, middleware, RTK Query, etc.	Need lightweight state management

★ 9. Middleware & DevTools

RTK	Zustand
✓ Built-in support ✓ Rtk Toolkit DevTools ✓ Middleware ecosystem	✓ Devtools middleware ✓ Persist middleware ✓ Very flexible

★ 10. Learning Curve

RTK
Medium

Zustand is easier for beginners.

Zustand
Easy

★ 11. Performance

Both are fast ⚡ but Zustand is slightly lighter (smaller bundle size).

★ 12. Feature Comparison

Feature	RTK	Zustand
State Management	Yes	Yes
Boilerplate	More	Very Less
Setup	Complex	Simple
DevTools	Built-in	Middleware
Middleware	Built-in support	Manual add
Async Logic (RTK Query)	Yes	No (Manual)
Persist State	Yes	Yes (Middleware)
Best For	Large & Complex Apps	Small & Medium Apps
Learning Curve	Medium	Easy
Flexibility	Structured	Highly Flexible

★ 13. When to use What?

Use RTK when:

- Building large scale app.
- Need strict structure.
- Need RTK Query for API.
- Working in a big team.

VS

Use Zustand when:

- Building small / medium app.
- Need simple global state.
- Want less boilerplate.
- Like flexible approach.

★ 14. Quick Notes

✓ RTK → Structure + Convention + Powerful.
Good for long-term large projects.

✓ Zustand → Simplicity + Flexibility + Lightweight.
Good for quick and modern apps.

✓ Both are great!
Choose based on project size and team preference.

★ 15. One Line Summary

RTK
More boilerplate,
More power,
Better for big apps.

VS

Zustand
Less code,
More flexibility,
Better for simple apps.

★ RIGHT TOOL, RIGHT TIME, RIGHT PROJECT ★